

Classical Pre-Oracle Filtering for Quantum Oracle Sketching via Wheel Constraints

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Abstract

Quantum Oracle Sketching (QOS) enables quantum-accessible representations of massive classical datasets without full storage. We introduce a classical pre-oracle filtering layer based on modular wheel constraints (mod30, mod210, mod2310) that reduces the candidate input stream before oracle construction. Across these filters, candidate streams are reduced by approximately 73–79%. Experiments on synthetic data and the 20 Newsgroups dataset show that substantial reductions in sample size preserve usable downstream classifier behavior while decreasing training time (used here as a simple workload proxy). These results support a tunable tradeoff between classical preprocessing and oracle construction cost, without modifying QOS internals.

1 Introduction

Quantum Oracle Sketching (QOS) provides a framework for constructing quantum-accessible representations of large classical datasets via sampling and sketching. A central challenge in such workflows is the scale of the classical input stream prior to oracle construction.

This work introduces a simple, non-invasive preprocessing layer: modular wheel-based filtering applied before QOS sampling. The goal is not to alter QOS algorithms, but to reduce the candidate stream entering oracle construction.

We study three wheel filters:

- mod30: $2 \cdot 3 \cdot 5$
- mod210: $2 \cdot 3 \cdot 5 \cdot 7$
- mod2310: $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$

This work complements the QOS framework by introducing a classical preprocessing layer that reduces the input stream prior to oracle construction.

Code Availability Code and experiments are available at: <https://github.com/thinkthoughts/quantum-oracle-sketching-mod30>. The modwheel filtering layer is developed in the branch: <https://github.com/thinkthoughts/quantum-oracle-sketching-mod30/tree/modwheel-layer>.

2 Wheel-Based Pre-Oracle Filtering

A wheel filter is defined by a modulus

$$M = \prod_i p_i,$$

where admissible residues satisfy

$$\gcd(r, M) = 1.$$

Wheel	Modulus	Residues	Candidate Fraction
mod30	30	8	0.2667
mod210	210	48	0.2286
mod2310	2310	480	0.2078

Table 1: Wheel filters and candidate fractions.

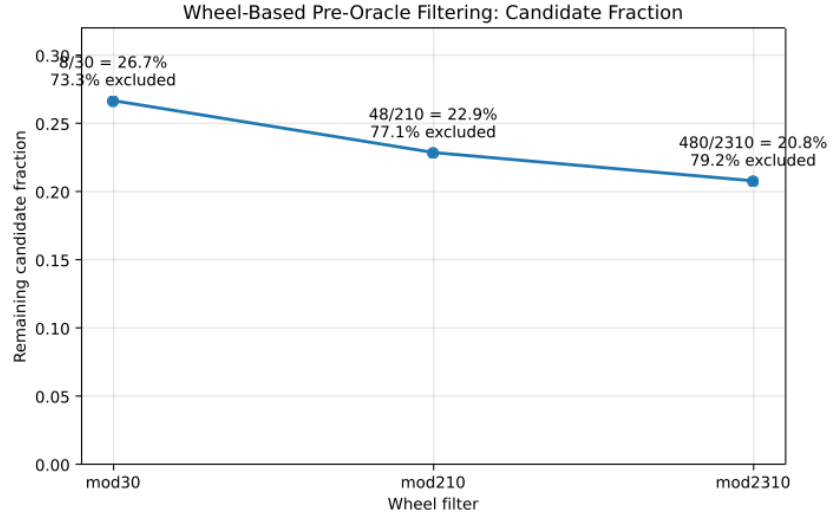


Figure 1: Wheel candidate fraction and reduction.

3 Adapter Design

X, y dataset
 -> row indices
 -> wheel filter
 -> filtered indices
 -> X_filtered, y_filtered
 -> downstream processing

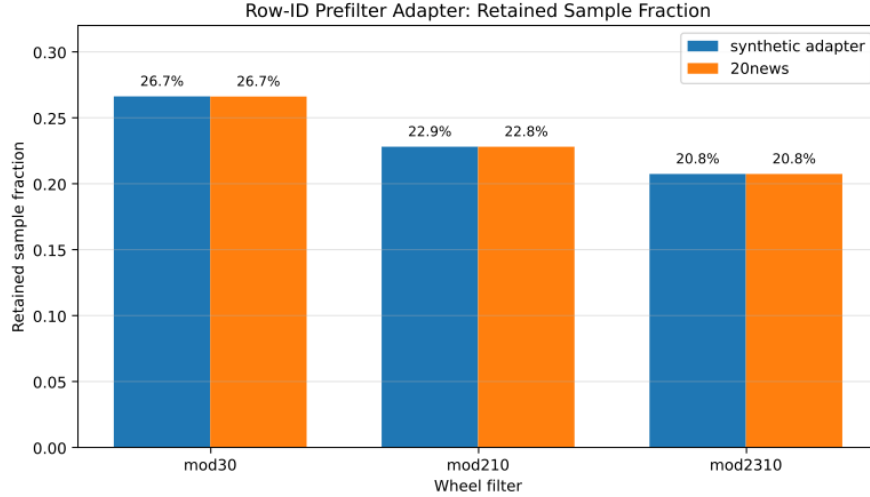


Figure 2: Retained sample fraction for synthetic adapter and 20news filtering.

Subset	Samples	Training Time (s)	Balanced Accuracy
baseline	3729	0.0478	0.854
mod30	994	0.0155	0.854
mod210	852	0.0130	0.877
mod2310	775	0.0127	0.840

Table 2: 20 Newsgroups row-ID filtering results.

4 Experiments

4.1 Synthetic Adapter

4.2 20 Newsgroups Dataset

5 Results

These results empirically support an adapter-level pipeline in which classical filtering reduces the input stream prior to downstream QOS or machine learning workflows.

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classical filtering
-> reduced input stream
-> downstream QOS / ML workflow
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6 Discussion

This work does not modify QOS internals, does not claim quantum advantage, and does not claim accuracy improvement.

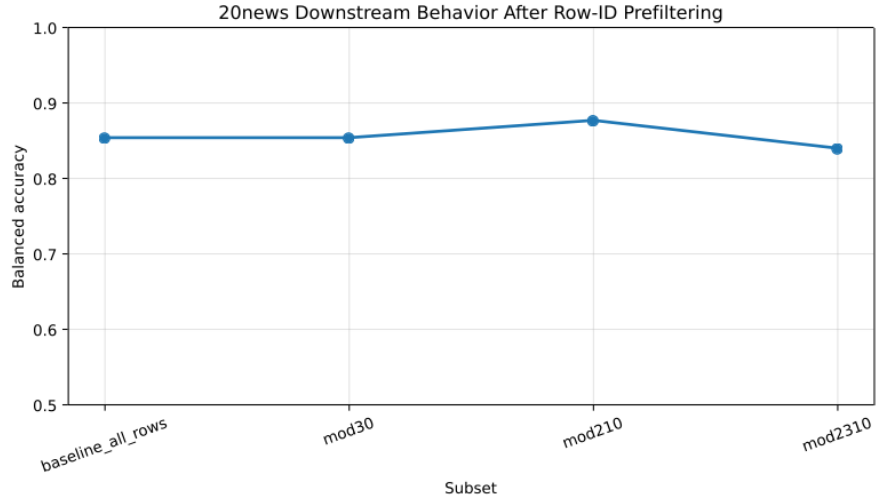


Figure 3: Balanced accuracy after filtering.

7 Conclusion

We introduced a classical preprocessing step that reduces candidate sets while preserving downstream behavior.

References

H. Zhao. *Exponential Quantum Advantage in Processing Massive Classical Data*. arXiv:2604.07639, 2026.

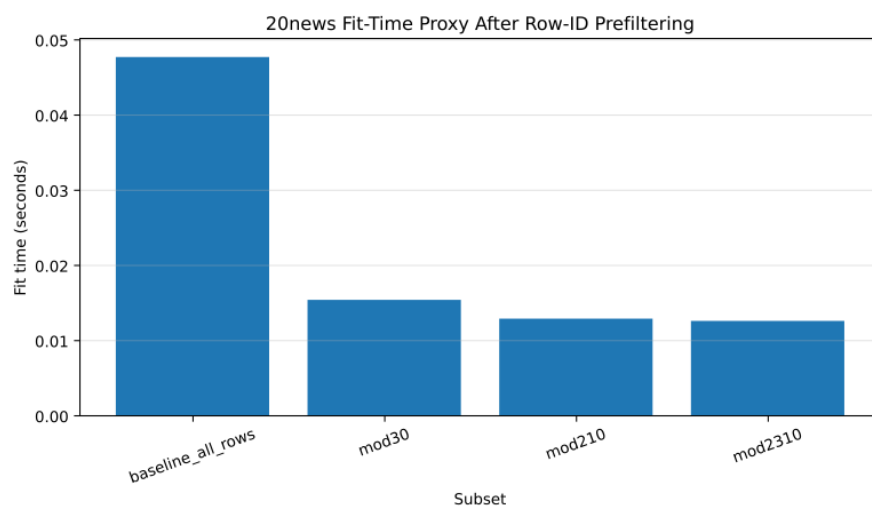


Figure 4: Training time proxy after filtering.